

VBF-T1R1NOFORCE-DS-01 - Cell Design Specification, virtual design

VBF-T1R1NOFORCE-DS-01 · Generation date: 2026-08-25 · Virtual design (simulation-based; no physical build)

1. Basic Specification

Field	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 baseline parameter set, pure-graphite negative)	`--base Chen2020`, parameter dump (no SiOx)
Nominal capacity	5.0 Ah (parameter set) / 5.654 Ah (simulated 1C, 25 °C)	`cell/r6_c7_1c_spme.json:capacity_ah`
Voltage window	2.5 – 4.2 V (charge upper cut-off 4.2 V; overcharge protocol = +0.5 V -> 4.7 V, matching task spec)	Chen2020 dump `Upper/Lower voltage cut-off`
Electrode geometry	0.065 m x 1.58 m = 0.1027 m² (unwound electrode, parameter-set geometry)	Chen2020 dump `Electrode height/width [m]`
Cell stack thickness	210.6 µm (75.6 + 95 + 16 + 12 + 12 µm); shell thickness: Not provided	`cell/r6_c7_energy.json:thickness_m`
Electrolyte formula	EC/EMC + LiPF6 (baseline) with transport overrides: conductivity 1.8 S/m, diffusivity 5e-10 m²/s (formulation-bridge #65, formula from simulation output)	`cell/r6_c7_params.json`
Cation transference	0.5 (override; baseline 0.2594)	`cell/r6_c7_params.json`; Chen2020 dump
Cooling design	Liquid cold plate, both pouch faces: h = 25 W/m²K, A = 2 x 0.1027 = 0.2054 m²	`cell/r6_c7_params.json`

2. Electrode and Separator

Layer	Material	Thickness	Porosity	Active fraction	Particle radius (design)
Positive	NMC811	75.6 µm	0.335	0.665	1.2 µm (single-crystal)
Separator	PP	12 µm	0.47	—	—
Negative	Graphite	95 µm (base 85.2 µm)	0.25	0.75	0.8 µm (fine graphite)
Positive CC	Al	16 µm	—	—	—
Negative CC	Cu	12 µm	—	—	—

N/P ratio (capacity basis) = (neg capacity density x thickness) ÷ (pos capacity density x thickness)
= (33133x0.75x96485/3600x95e-6) ÷ (63104x0.665x96485/3600x75.6e-6) = 63.280 ÷ 85.020 = **0.744**.

Source: Chen2020 dump (`Maximum concentration`, active fractions) + override thickness. Note: this baseline parameterization is anode-limited (capacity N/P < 1); the measured 1C capacity (5.654 Ah) tracks the anode inventory — verified in R4 (capacity grew linearly with anode thickness).

3. Process Design Parameters

Parameter	Formula	Value	Unit
Positive areal density	thickness x (1-porosity) x density	163.99	g/m²
Negative areal density	thickness x (1-porosity) x density	118.06	g/m²
Positive compaction density	areal density ÷ thickness	2.169	g/cm³
Negative compaction density	areal density ÷ thickness	1.243	g/cm³
Electrolyte fill amount	pore volume x 1.2 g/cm³ (literature density) x fill factor 1.0	6.74	g/cell
Formation recommendation	design value: 0.1C CC to 4.2 V, 25 °C, 2 cycles (production-line value requires tuning)	—	—

Areal densities sourced from `cell/r6\_c7\_energy.json:layer\_kg\_m2` (contract caliber). Electrolyte density is a literature value (1.2 g/cm³), annotated as such.

4. Mass Breakdown

Layer	kg/m²	g/cell	Source
Positive electrode	0.163994	16.842	`cell/r6_c7_energy.json:layer_kg_m2` x area
Negative electrode	0.118061	12.125	same

Layer	kg/m ²	g/cell	Source
Positive CC (Al)	0.043200	4.437	same
Negative CC (Cu)	0.107520	11.042	same
Separator	0.002525	0.259	same
Total (contract, electrolyte excluded)	—	**44.705**	`cell/r6_c7_energy.json:mass_kg`
Electrolyte (estimate)	—	6.743	pore volume x 1.2 g/cm ³ (literature)
Total incl. electrolyte (estimate)	—	**51.448**	mechanical sum

5. Performance Verification (vs entry-0 criteria)

Item	Result	Criterion	Verdict
Energy density	457.93 Wh/kg	>= 392.61 Wh/kg	[PASS] PASS (+65.3)
4C charge @45 °C, max temperature	318.79 K	<= 333.15 K (60 °C)	[PASS] PASS (margin 14.4 K)
4C charge lithium plating	anode potential min +0.00824 V (SPMe); +0.00515 V (DFN cross-check)	checked < 0 V	[PASS] PASS
Overcharge to 4.7 V thermal runaway triggered	= false, T_max 298.22 K	no thermal runaway	[PASS] PASS
1C discharge capacity	5.654 Ah	(no task threshold — informational)	—
4C CC charge acceptance	4.63 Ah = 81.9 % SOC in ~836 s at 20 A	(design-quality gate ~80 %)	[PASS] met

Sources: `cell/r6\_c7\_energy.json`, `cell/r6\_c7\_4c.json`, `cell/r6\_c7\_4c\_dfn.json`, `cell/r6\_c7\_oc.json`, `validation/r6\_c7\_tr.json`.

6. Design Notes (what changed and why)

- Cooling**: parameter-set cooling area (0.00531 m²) is an edge-cooling artifact for a 0.1027 m² pouch; double-face cold-plate cooling (0.2054 m², h = 25 W/m²K) is the standard pack design. This solved T_max (354.3 -> 318.8 K). (R2->C3)
- Electrolyte transport**: sigma 1.8 S/m, D 5e-10 m²/s, t+ 0.5 (formulation-bridge estimates) cut charge-onset polarization. (R2->C2)
- Particle size balance**: cathode 1.2 µm single-crystal (tau_solid = R²/D = 360 s << 900 s 4C window; commercial practice); anode 0.8 µm fine graphite — the anode kinetic cut was the decisive plating lever (+16 mV for 1.5->1.0 µm, +7 mV more for 1.0->0.8 µm). (R5->C7)
- Anode thickness 95 µm**: baseline is anode-limited (capacity N/P 0.744); thickening added capacity (+11.4 %) and energy without ED penalty (400.8 -> 456–458 Wh/kg). (R4)
- Honest caveats**: plating margin is thin (+5 mV under DFN) — prototype 4C validation recommended before production; the 4C charge-side `capacity\_ah` in runner outputs under-reports by the nominal-capacity factor (x5 for Chen2020) — true charge capacity derived mechanically (0.92596 x 5 = 4.63 Ah) and noted per round.